Rahul Pandare

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EDUCATION

Institute of Chemical Technology, Mumbai ***2015-2019***

Bachelor of Chemical Engineering with Cumulative Grade Point Average: 7.58/10

Pace junior science college, Mumbai ***March 2015***

Higher Secondary Certificate (H.S.C.) Examination: 88.15%. Awarded the Scholarship for Higher Education (SHE) for being in the top 1% of the school board at class XII level.

PROFESSIONAL EXPERIENCE

Aker Solutions, Mumbai ***August 2019 – Present***

Process Engineer (Trainee)

Worked on several EPCC projects such as 0.9 MMTPA Ethylene capacity Dual Feed Cracker Unit (DFCU) project, Petrochemical Fluid Catalytic Cracking Unit (PFFCU) pre-bid project and Sonatrach's gas field extension project:

- Involved in preparation of engineering documents like Equipment datasheets and Piping and instrumentation diagram. Owing to a good grasp on the process, suggested several cost-effective process specific modifications to our client.
- Performed activities such as line sizing, vessel sizing, pressure drop calculations, pump hydraulics and relief valve calculations.
- Significantly contributed to the projects by preparing several excel based tools and standard calculation spreadsheets essentially resulted in saving several man-hours.
- Performed thermal rating for various types of shell and tube heat exchangers on HTRI software. Familiar with various APIs such as API-520, API-521 and API 620.
- Compared the performance feasibility of two available options for the de-carbonization unit for the gas field extension project, namely Amine absorption system and Membrane absorption system for natural gas. Considering the numerous parameters and perusing the existing literature on the same made an informed suggestion to consider former over the latter.

Institute of Chemical Technology, Mumbai ***May 2017-May 2019***

Undergraduate Research Assistant

Worked under the guidance of prof. P.K. Ghosh from the Chemical engineering department at ICT and alongside his post-doctoral student Dr. Lokeshkumar Ramteke to tackle an industrial problem:

- Developed a chemical process to recycle dimethyl-glyoxime (DMG), an expensive reagent, used in removal of nickel impurities from industrial zinc sulfate solutions.
- Optimized the above chemical system to increase reaction yields and decrease operating costs by approximately 12% each.
- Provided consultations to Rubamin Ltd., a zinc product manufacturing company based in Vadodara, India, to reduce nickel impurities in their product stream and thus improve product quality.

Panorama Consulting Engineers India, Mumbai ***May 2018 – August 2018***

Process engineering intern

Worked on a Greenfield project to conceptualize a batch process into a continuous process:

- Developed a scheme of continuous operation to produce Methoxyamine. Understanding the kinetics of the system performed the mass balance, heat balance, sizing and design calculations for process equipments such as PFR, venturi tube and flash drum.
- Provided support in preparing P&IDs using AutoCAD and brainstormed on several process development issues.

ACADEMIC PROJECTS AND PRESENTATIONS:

Homepaper on '*Graphene synthesis and application of Graphene derivatives on Textiles'* under **Prof. Ravindra Kale**, Dept. of Fibres & Textile Processing Technology, ICT - Mumbai and **Prof. K.V. Marathe**, Dept. of Chemical Engineering, ICT – Mumbai. (August 2018 – May 2019)

- Synthesized Chlorographene and studied the application of Chlorographene on Polyester fabric which yields electrical conduction properties.
- Performed an exhaustive literature survey for selecting the process route and gathering the essential data.
- Estimated the rough economics of the selected process system to check the feasibility of its scale-up.

Seminar on '*Chemical Systems in synthesis and separation requiring population balance approach'* under **Prof. A.B. Pandit**, Dept. of Chemical Engineering, ICT – Mumbai. (November 2018)

- Discussed the use of population balance equations (PBEs) in different systems and reviewed how PBEs are formulated and possibly solved using two case studies of Crystallization and Granulation.

Project on '*Optimization of solution flowrate for adsorption in a fixed bed* ' for the course of Optimization of Chemical Engineering Systems under **Prof. V. H. Dalvi**, Dept. of Chemical Engineering, ICT – Mumbai. (December 2018)

Studied the column dynamics for adsorption by analyzing the breakthrough curves of the system at various parameters to arrive at the optimum flow rate of the solution.

- The mass balance equation was solved using the PDE solver function on MATLAB using the initial-boundary value. The solver converts the partial differential equations to ordinary differential equations using a second-order accurate spatial discretization, based on a fixed set of user specified nodes. After discretization, differential equations give rise to algebraic equations.
- Axial diffusion and mixing at the exit of the deposit is determined through solving Dancwart's boundary conditions. The mass transfer coefficient is estimated by the Ranz & Marshal correlation.
- Adsorption system considered: Ginsenoside as solute and Amberlite XAD-7 resin particles as adsorbent. Properties of solute, adsorbent, and bed were obtained from literature.

EXTRA CURRICULAR AND ACHIEVEMENTS

- Organized a Heath, Safety, Security and Environment (HSSE) quiz to create awareness among the employees as a part of Solutioneers, an international employee network that brings together young talent for professional development, networking and fun. (2019)
- 2nd place in 'Junior IDP' (Flagship event) at Vortex, the Largest Chemfest in India (2015)
- Cleared JEE-MAIN 2015 with 95.8 %ile out of 1,500,000 students. (2015)
- Awarded the Scholarship for Higher Education (SHE) under innovation in science pursuit for Inspired Research (INSPIRE) by virtue of performance within top 1% of the school board at class XII level in the examination held during the academic session ending March 2015.
- Served as the Graphic Design Head (Core Committee) for Vortex-2017, the Largest Chemfest in India. Efficiently managed a team of 12 people to deliver the graphic design needs of our event managers, sponsors and public relations team.

TECHNICAL SKILLS

- **Programming languages**: C, C++, MATLAB, Python 3.8 – conversant with various libraries like Pandas, Numpy, Seaborn and Beautiful soup.
- **Design and Simulations software**: SolidWorks, HTRI
- **Graphic design software**: CorelDraw, Photoshop, Adobe After Effects.

LANGUAGES

English, Hindi, Marathi, Spanish (A1), French (8th to 10th grade)